

MOHAMMED AKROUR

AI Engineer

Paris, France | +33 7 80 79 66 07 | akroummohammed@gmail.com | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

PROFILE

AI Engineer with a strong foundation in applied mathematics, machine learning and generative AI, experienced in taking AI systems from prototype to production. Skilled in building LLM-based applications (RAG, multi-agent systems), scalable backend APIs and full ML lifecycles (Python, FastAPI, Docker, CI/CD), with rigorous evaluation and monitoring practices. Curious, autonomous and a clear communicator; fluent in French with professional English.

TECHNICAL SKILLS

Generative AI & NLP: LLMs, Retrieval-Augmented Generation (RAG), LangGraph (StateGraph, multi-agent orchestration, tool calling, human-in-the-loop, prebuilt tools), agentic AI / multi-agent systems, Model Context Protocol (MCP), LangChain, conversational AI, prompt engineering, embeddings, semantic search, vector databases (Qdrant, ChromaDB), Hugging Face, LLM evaluation (LangSmith), hallucination mitigation

Machine & Deep Learning: scikit-learn, PyTorch, TensorFlow, XGBoost, LSTM, statistical modeling, predictive analytics, feature engineering, anomaly detection, time-series analysis, model evaluation, explainable AI (SHAP)

Software & MLOps: Python, FastAPI, Flask, REST APIs, JavaScript, React, Docker, Git/GitHub, CI/CD, model deployment, monitoring, LLMOps, Google Cloud (GCP)

Data: Pandas, NumPy, SQL (PostgreSQL, MySQL), data ingestion, data pipelines, ETL, data processing

EXPERIENCE

AI Engineer — End-of-Studies Internship (Master's)

Jan 2025 – Jun 2025

Université Paris-Est Créteil (UPEC), Paris — Supervisor: Prof. Amine Nait Ali

- Designed and deployed a production-grade AI system (FastAPI, Docker, React) for processing heterogeneous health data from APIs, time-series signals, and PDF documents (OCR).
- Built and integrated LLM-based pipelines (RAG) into scalable backend APIs, enabling automated exploration of structured and unstructured data and generation of analytical reports.
- Implemented semantic retrieval workflows using embeddings and vector databases, improving search relevance and response accuracy across large document collections.
- Designed evaluation and validation workflows for LLM outputs, identifying failure cases and improving answer quality to reduce hallucination risks.
- Developed robust data pipelines for ingestion, transformation and analysis, ensuring reliability across diverse data sources.
- Developed a hybrid disease-risk prediction system combining an LSTM Autoencoder and XGBoost, providing explainable predictions (SHAP) over temporal patient data.

Gas Turbine Maintenance Study — Internship

Feb 2022 – Jun 2022

Sidi R'cine Refinery, Sonatrach — Algeria

- Analyzed industrial sensor data (temperature, pressure, vibration) within an operational refinery of the Sonatrach group.
- Contributed to predictive maintenance projects based on time-series analysis, applicable to equipment monitoring and measurement data analysis.
- Strengthened analytical rigor and measurement precision in a demanding industrial and petroleum environment.

SELECTED PROJECTS

Multi-Agent Research Assistant — LangGraph, LangChain, Gemini, Tavily, FastAPI, Docker

- Built an autonomous multi-agent system (Planner, Searcher, Writer, Critic) in a self-correcting LangGraph loop that researches any topic and revises its answer over up to 3 iterations.
- Orchestrated real-time web retrieval (Tavily) and LLM synthesis with tool calling, deployed via FastAPI and Docker and monitored with LangSmith.

RAG Document Q&A System — Python, LangChain, Qdrant, Gemini, FastAPI

- Designed a production-grade RAG pipeline enabling users to upload documents and query them via a REST API, covering ingestion, retrieval and answer generation.
- Built modular LLM workflows combining dense vector search (Qdrant) and BM25 keyword retrieval with cross-encoder re-ranking to improve answer relevance and precision.
- Implemented evaluation pipelines (faithfulness, answer relevancy, context recall) to systematically assess and improve LLM output.
- Identified and mitigated retrieval and reasoning failure cases, improving robustness and reducing incorrect or hallucinated responses.
- Containerized the system with Docker and exposed endpoints via FastAPI, enabling scalable and maintainable deployment.

Document AI Pipeline — OCR & Multilingual NER — Tesseract, YOLO, LayoutLMv3, mBERT

- Built a full-stack Python application to process scanned and digital documents, combining OCR (Tesseract), YOLO layout detection and LayoutLMv3 for automated structured information extraction.
- Engineered data pipelines for ingestion, cleaning and transformation of heterogeneous document data, ensuring reproducibility in production-like workflows.
- Fine-tuned mBERT for multilingual Named Entity Recognition (EN/FR) with BIO tagging, evaluated with precision, recall and F1-score.

MLOps Churn Prediction Platform — End-to-End ML Lifecycle — MLflow, FastAPI, Docker, GitHub Actions, Evidently AI

- Built an end-to-end MLOps pipeline for customer-churn prediction with experiment tracking and a model registry (MLflow) and automated CI/CD (GitHub Actions).
- Served the XGBoost model via a Dockerized FastAPI REST API, deployed live on the cloud with health and prediction endpoints.
- Implemented data-drift monitoring (Evidently AI) with an automated retraining trigger, validating the loop on simulated production data shift.

Parameter-Efficient Fine-Tuning of GPT-2 (LoRA) — PyTorch

- Implemented supervised fine-tuning of GPT-2 using LoRA with a custom PyTorch training loop (causal LM loss, gradient accumulation, learning-rate scheduling).
- Designed evaluation workflows to compare fine-tuned models with baseline prompting, reducing training loss.

Anomaly Detection on Sensor Time-Series — scikit-learn, PyTorch

- Developed and evaluated ML models (Isolation Forest, SVM, KNN, BiLSTM) for anomaly detection in multivariate time-series data.

EDUCATION

M.Sc. in Biometrics & Intelligent Vision — Université Paris-Est Créteil (UPEC), France

2023 – 2025

M.Sc. in Mechatronics — Université M'Hamed Bougara de Boumerdès (UMBB), Algeria

2020 – 2022

LANGUAGES

French — Fluent English — Professional proficiency Arabic — Native